WAIMLAp 2026

Workshop on Artificial Intelligence and Machine Learning Applications

PUCP

Frozen DINOv3 Features for Martian Terrain Segmentation: Layer-wise Linear Probing on AI4Mars

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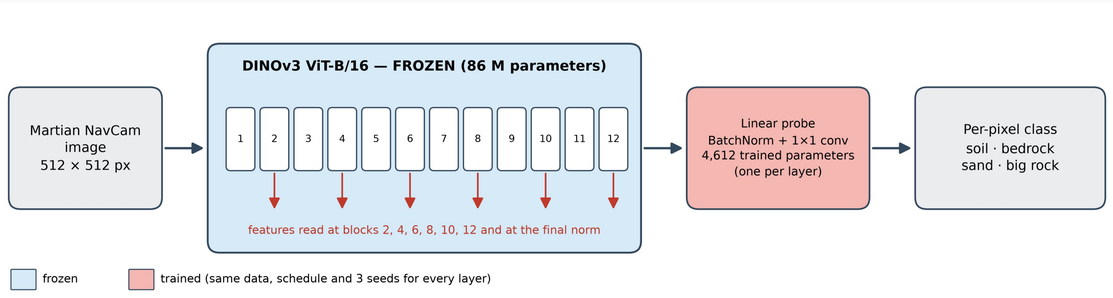
MOTIVATION & RESEARCH PROBLEM

- Rovers need to recognize the terrain they drive on (soil, bedrock, sand, big rock); expert labels of Mars are scarce and costly.
- DINOv3 is pretrained on web images, far from Mars: grayscale, repetitive textures, unusual illumination.
- **Question:** how well do frozen DINOv3 features transfer to Martian terrain, and which layers carry the useful information?

OBJECTIVES / CONTRIBUTION

- Measure the transfer of frozen DINOv3 ViT-B/16 to AI4Mars (Curiosity): layer-wise linear probe (4,612 trainable parameters), then a larger decoder and partial fine-tuning.
- Uncertainty: 3 seeds and bootstrap confidence intervals over images; six hypotheses and decision rules fixed before the final run.
- Controls: random-init ViT, DINOv2, label efficiency, label-border noise and published baselines (M1–M3).
- None of the seven AI4Mars baseline papers we reviewed uses a frozen foundation backbone (MANTLE [7] does, for binary boulder segmentation on its own MSL data).

METHODOLOGY



- **Data:** AI4Mars, Curiosity NavCam, 512×512 px. 15,564 training images (crowd labels), 500 validation (temporal hold-out), 322 expert test images in three variants M1 / M2 / M3, scored at native label resolution.
- **Model:** DINOv3 ViT-B/16, frozen; features read at blocks 2, 4, 6, 8, 10, 12 and the final norm.
- **Heads:** linear probe (BatchNorm + 1×1 conv, 4.6 k parameters) per layer; decoder fusing blocks 3, 6, 9 and 12 (3.5 M); last 2 or 4 blocks fine-tuned (14 M / 28 M). 3 seeds each.
- **Evaluation:** mIoU over 4 classes; layer and epoch chosen on validation, the test set only confirms; 95 % paired bootstrap CIs over images; a difference counts as real if its CI excludes 0 and $|\Delta| \geq 0.01$.

WHERE IT FAILS

- **Big rock is the hardest class:** validation IoU 0.24 (probe) vs 0.62–0.88 for the others, and 0.13 / 0.12 on test M1 / M2; 69 % of its pixels are predicted as bedrock (probe), 60 % (decoder).
- **M3 hides this:** big rock appears in only 5 of its 322 images (77,698 px). Every model finds 96–99 % of those pixels; they differ in false alarms (probe 109 k px, decoder 16 k px), which drives the M3 mIoU gap (0.83 vs 0.93). Without big rock all models reach 0.96–0.97.
- Size alone does not explain the misses: recall is 0.28 for the smallest third of big-rock regions and 0.34 for the largest (inconclusive).

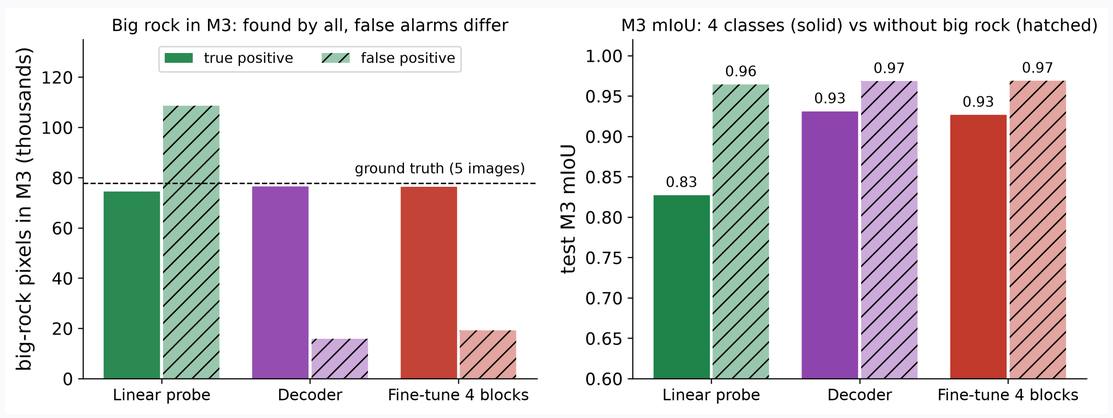


Fig. 7. Big rock in test M3 (mean of 3 seeds): true and false positives (left); mIoU with and without that class (right).

CONCLUSIONS

- **Frozen DINOv3 features transfer to Mars** without adapting the backbone: a linear probe (4.6 k trainable parameters) is within 0.025 mIoU of the best published end-to-end model on M1 and M2, and heavier heads add at most 0.012.
- The gain comes from pretraining (+0.41 over a random ViT, +0.013 over DINOv2) and the final layer is as good as any other: no intermediate layer helps.
- M1/M2 saturate early: 250 labelled images give 99 % of the probe's mIoU, and removing label borders raises every head by ≈ 0.025, so part of the ceiling may be label noise (not conclusive).
- **Limits:** one backbone, one rover, 3 seeds, 322 test images per variant; only 1 of 6 pre-specified hypotheses was supported, the rest are inconclusive at this sample size; M3 cannot rank models on big rock; the baseline comparison is indicative.
- **Next:** retrain a published baseline under the same protocol, other rovers or datasets, big-rock-oriented losses.

RESULTS & DISCUSSION

- **Transfer:** a linear probe on the final layer reaches mIoU 0.677 / 0.725 on test M1 / M2 (0.651 on validation; 0.125 for always predicting the majority class).
- **Heavier heads add little:** decoder 0.684 / 0.735, last 4 blocks fine-tuned 0.687 / 0.737 — at most +0.012 over the probe (decoder minus probe on M1: +0.007, 95 % CI [−0.001, +0.019]).
- **Pretraining matters:** a random-init ViT-B/16 gets 0.264 on M1 (DINOv3: +0.413); DINOv2 B/14 gets 0.664 and DINOv3 is +0.013 [+0.002, +0.025] on M1 but not clearly on M2; matching the token grid (DINOv2 at 448 px) leaves the gap unchanged (+0.014 [+0.003, +0.026]).
- **Baselines:** the probe is level with the 512-px published models and 0.023 / 0.015 below the best (UPerNet, 1,024 px) on M1 / M2; fine-tuning the last 4 blocks is 0.013 / 0.003 below, with 0.005–28 M trainable parameters instead of 47–81 M. Indicative only: published values are single runs on another split.
- **Layers:** mIoU rises with depth and block 6 already gives 98 % of the final M1 value; no intermediate layer beats the last one (block 10, best on validation: −0.010, 95 % CI [−0.021, +0.001]).
- **Labels and borders:** 250 images give 99 % of the 4,000-image probe mIoU on M1; excluding an 8 px band around label edges raises every head by ≈ 0.025 (probe 0.677 → 0.702), consistent with, but not proof of, label noise at borders.

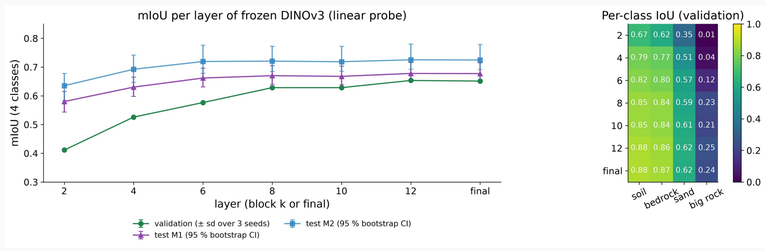


Fig. 1. mIoU per layer of frozen DINOv3 (validation ± sd over 3 seeds; test sets with 95 % bootstrap CI) and per-class IoU on validation.

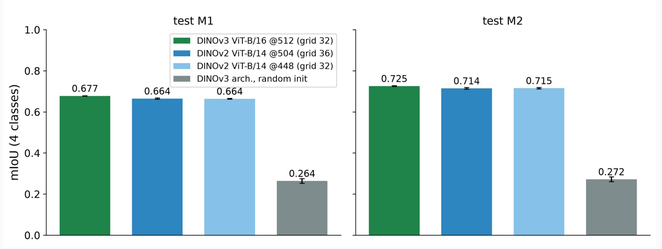


Fig. 2. Frozen backbones, same probe (DINOv3, DINOv2 at two input sizes, random init).

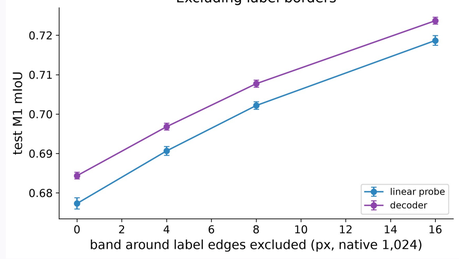


Fig. 3. Excluding label borders (M1).

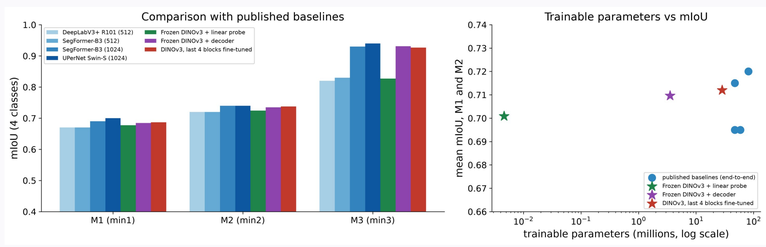


Fig. 4. Comparison with published baselines (M1–M3; M3 is dominated by big-rock false alarms) and trainable parameters.

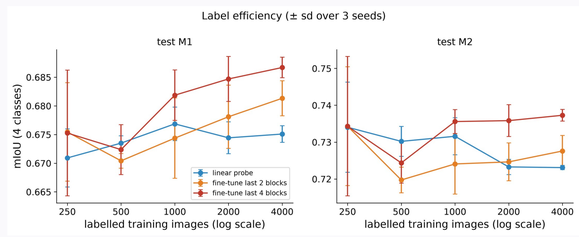
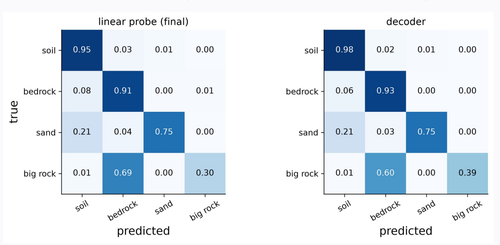


Fig. 5. Label efficiency: linear probe vs fine-tuning the last 2 or 4 blocks (± sd over 3 seeds).



Main confusions (validation, row-normalised, probe / decoder):

big rock → bedrock 68.6 % / 60.0 %

sand → soil 20.5 % / 21.3 %

sand → bedrock 4.2 % / 3.4 %

Fig. 6. Row-normalised confusion matrices on validation: linear probe on the final layer (left) and decoder (right).

KEY REFERENCES

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[6] E. Goh, J. Chen, B. Wilson, “Mars terrain segmentation with less labels,” arXiv:2202.00791, 2022.
[7] P. Durai, G. Doran, “MANTLE: a framework for adaptive in-situ planetary perception using a modular uplink principle,” arXiv:2608.28724, 2026.

CONTINUE THE CONVERSATION

Code, notebook and results: https://github.com/gonzaquis29/DinoV3-AI4Mars

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Data and model: AI4Mars (NASA/JPL), DINOv3 (Meta AI). Compute: free-tier Kaggle GPU.

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